

Bryan Torres

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EDUCATION

Virginia Polytechnic Institute and State University (Virginia Tech) <i>Master of Engineering in Computer Science & Applications</i> GPA: 3.90 / 4.00 Focus: Computer Systems & Architecture, Software Systems Relevant Coursework: Data Structures & Algorithms, Operating Systems, Machine Learning, Software Engineering	Blacksburg, Virginia Expected May 2027
Virginia Polytechnic Institute and State University (Virginia Tech) <i>Bachelor of Science in Computer Science</i>	Blacksburg, Virginia Aug. 2022 – May 2026

EXPERIENCE

Software Engineer Intern Expedition Technology <i>Geospatial Dataset Diversity</i>	May 2026 – Aug. 2026 <i>Herndon, Virginia</i>
- Built and shipped Python/FastAPI backend APIs and query logic to compute 0–1 diversity scores across large-scale geospatial imagery datasets using normalized Shannon entropy over geographic, temporal, illumination, and image-quality metadata. - Delivered the feature end-to-end by integrating backend APIs with a TypeScript/Vite frontend and building Chart.js and GeoJSON/TopoJSON visualizations to surface dataset coverage gaps and distribution patterns. - Improved deployment reliability by extending reusable Helm/Argo CD CI/CD templates to automate pre-deployment database migrations for Kubernetes-based services.	
Software & DevOps Engineer Intern Dark Wolf Solutions <i>Project Syndicate</i>	Jun. 2025 – Aug. 2025 <i>Herndon, Virginia</i>
- Built an end-to-end cloud-native CI/CD security platform spanning four containerized services across a Next.js frontend, FastAPI backend, PostgreSQL data layer and Dex/OIDC authentication. - Deployed and managed Next.js and FastAPI services on Google Kubernetes Engine using Docker, Kubernetes, and Argo CD; integrated PostgreSQL persistence and Dex/OIDC identity infrastructure. - Implemented GitLab CI security gates with SonarQube and OWASP checks and built Jest/PyTest test suites, raising overall test coverage to 80%; reduced container image size by 22%.	
Undergraduate Researcher Virginia Tech <i>MARCO – Multi-Agent Code Optimization for HPC</i>	Oct. 2024 – Dec. 2025 <i>Blacksburg, Virginia</i>
- Extended MARCO, a multi-agent AI code optimizer, to CUDA/HPC workloads using GPU-library templates and iterative compile/run/verify feedback. - Built benchmarking infrastructure and ran NVIDIA V100 experiments comparing MARCO with HPC-Coder and CUDA baselines; MARCO achieved $2.1\times$ and $9.7\times$ higher GFLOPS at 512 and 1024 GEMM+ReLU matrix sizes. - Evaluated reference-guided GEMM optimization against CUTLASS; MARCO-generated kernels matched reference performance within approximately 1.3% across 512–2048 matrix sizes. Publication: Co-authored <i>MARCO: Multi-Agent Code Optimization with Real-Time Knowledge Integration for High-Performance Computing</i> [arXiv] (2025).	

TECHNICAL SKILLS

- Languages:** Python, TypeScript, JavaScript, C, Java, C#
- Frameworks & Data:** React, FastAPI, Next.js, Vite, Node.js, PostgreSQL, SQLAlchemy, Alembic
- Cloud & DevOps:** Docker, Kubernetes, Helm, Argo CD, GitLab CI/CD, Google Cloud Platform, SonarQube
- AI, Testing & Tools:** OpenAI/Anthropic APIs, Hugging Face, PyTest, Jest, CUDA, CUTLASS, Git/GitHub, Linux/Unix